

Extraction of aflatoxins by using mesoporous silica (type UVM-7), and their quantitation by HPLC-MS

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Abstract

A solid-phase extraction procedure has been developed by using a sorbent derived from UVM-7 mesoporous silica. The sorbent was applied to the extraction of aflatoxins B₁, B₂, G₁ and G₂ from tea samples followed by HPLC with mass spectrometric detection. The sorbent was characterized by transmission electron microscopy, nuclear magnetic resonance, X-ray diffraction and nitrogen adsorption-desorption. Extracted UVM-7 is found to be the best solid phase. The amount of solid-phase, type and volume of eluent, pH value and ionic strength and breakthrough volume were optimized. Following the recommended procedure, recoveries between 96.0 and 98.2% were achieved, with RSD values of <5.1%, and the limits of detection are in the range from 0.14 to 0.7 µg·kg⁻¹. The material is reusable. The method was applied to the analysis of real tea samples. A low matrix effect is found, and recoveries are >88%. The results were compared with those obtained by immunoaffinity columns as a reference method. Only low concentrations of aflatoxin G₂ were found in some samples, and results obtained with both methods are shown to be statistically sound and comparable.

Keywords

Tea, Preconcentration, Mycotoxin, Materials, Food analysis, Solid-phase extraction, Nanomaterials, Hierarchical porosity

Introduction

Mycotoxins are metabolites produced by fungi contaminating various food and feed crops, being a risk for both human and animal health [1]. Hence, mycotoxin contamination is an ongoing global concern, as it is considered an unavoidable and unpredictable problem, even where good agricultural, storage, and processing practices are implemented [2]. In fact, according to the Food and Agriculture Organization of the United Nations, 25% of the world's crop are contaminated with mycotoxins during growth or storage. Among them, the most commonly present naturally mycotoxins are aflatoxins (AF), mainly produced by *Aspergillus flavus* and *Aspergillus parasiticus*. Until now, eighteen different types of AFs have been identified, wherein aflatoxin B₁ (AFB₁), B₂ (AFB₂), G₁ (AFG₁), and G₂ (AFG₂) are the most toxic, since they have been classified as Group 1 carcinogens by the International Agency for Research Cancer [1, 3].

Even though these compounds can be found in several kinds of food, their presence is especially frequent in vegetable products such as nuts, cereals, corn, rice, peanuts, oil seeds, and certain spices, as well as products processed and derived from them. Thus, the European Commission regulation [4] sets specific limits for these kinds of products, being these in the range 4-15 µg·kg⁻¹ for total AFs and 2-12 µg·kg⁻¹ for AFB₁ [5]. Nevertheless, some studies have found aflatoxins in several other food products not specified in this regulation, such as beer [6–9], jelly, tea [10–12], and other medicinal herbs [13–16]. In these cases, in Spain [17] the limits are set in 10 µg·kg⁻¹ for total AFs and 5 µg·kg⁻¹ for AFB₁ for food products in general [5]. In this sense, minor aflatoxin concentrations have been found in nutraceutical products such as tea or medicinal herbs, although their control is advisable due to their current high consumption [11].

Regarding to aflatoxin determination in food samples, several instrumental techniques have been applied, being the most common high performance liquid chromatography (HPLC), mainly coupled with a fluorescence detector (FLD) or mass spectrometry detector (MS). However, their low concentrations and complex matrices make advisable the use of sample treatment and clean-up techniques. In this sense, apart from techniques based on thin layer chromatography (TLC) and other screening methods, immunoaffinity columns (IAC) coating pre-treatment is the most common and reliable method. Due to its selectivity and minimum level of interfering matrix components [18], it is recommended by several official organizations such as the AOAC [19]. However, IAC method requires very expensive and disposable cartridges, which are generally designed for only one type of toxin, reducing the method multiresiduality [18]. Thus, several alternatives have been proposed for AF extraction and preconcentration,

being the most common those based on solid-phase extraction (SPE). Due to the interest in the development of new sorbents for SPE some alternatives have been proposed for immunosorbents, such as molecularly imprinted polymers (MIPs) or oligosorbents (OS) [18, 20]. Therefore, there is a special interest in developing new SPE sorbents for this purpose.

In this regard, large surface materials, and mainly mesoporous silica materials are especially interesting in a diversity of applications due to their favourable properties, such as stability low density, good biocompatibility or ease for control pore size and morphology [21, 22]. These singular features and their large surface areas (up to 1000 m²·g⁻¹) permit the adsorption of both small and large molecules by size-exclusion mechanisms. In addition, their wide range of post-modification functionalities, makes these materials attractive as SPE sorbents [23].

Although these materials have been known for a long time, the research on porous materials chemistry was increased from the discovery of the M41S silicas. Through the atrane route, the introduction of surfactants as “templates” permitted the control in the mesopore range compared to the classical sol-gel process, thus expanding their range of application [24]. In this way, UVM-7 silicas can be considered as a nanometric version of the well-known MCM-41 materials, being their main feature their particular architecture. It consists in a continuous network constructed from aggregated mesoporous nanoparticles generating highly accessible pore system with high surface area [25]. In this sense, UVM-7 materials have hardly been applied as sorbents in few studies, most of them focused on environmental analysis, using several modifications of these materials for metallic ions [26] or organic pollutants [27] determination. In addition, with regard to food analysis, few studies have been developed with UVM-7 mesoporous silicas for the retention of large components as phospholipids [28].

The aim of this work is to study the potential use of UVM-7 mesoporous silica as SPE sorbent for extraction and preconcentration of aflatoxins in nutraceutical products and their later analysis by HPLC-FLD and UHPLC-MS/MS. Relevant features of the method (such as repeatability, breakthrough volume, reusability, etc.) were established and the optimal SPE procedure was then applied to the preconcentration of aflatoxins in tea samples. Additionally, these results were compared with those obtained with the reference method of the AOAC, using commercial IAC.

Materials and methods

Chemicals and reagents

Aflatoxin standards were purchased from LGC Standards (www.lgcstandards.com, Teddington, Middlesex, UK), as a multicomponent solution in acetonitrile of concentration $2\ \mu\text{g}\cdot\text{mL}^{-1}$ for B₁ and G₁, and $0.5\ \mu\text{g}\cdot\text{mL}^{-1}$ for B₂ and G₂. Working standard solutions were prepared by dilution of this mixture and stored in darkness at -18 °C. Methanol (MeOH), ethanol and acetonitrile (ACN) from HPLC grade quality were used in extraction and analysis procedures, and were purchased from Panreac AppliChem (www.itwreagents.com, Darmstadt, Germany). Ultrapure water from Adrona (www.adrona.lv, Riga, Latvia) purification system was also employed.

For synthesis of mesoporous silica materials, cetyl trimethylammonium (CTAB) reagent grade $\geq 99\%$ from Sigma Aldrich (www.sigmaaldrich.com, St. Louis, USA) and ethanol from VWR ProLabo Chemicals (www.vwr.com, Radnor, USA) was employed, as well as tetraethyl orthosilicate $\geq 98\%$ (TEOS) and triethanolamine $\geq 99\%$ (TEA), both in reagent grade, from Fluka (www.lab-honeywell.com, Buchs, Switzerland).

In addition, for method validation with commercial IAC, Supel™ Tox AflaZea cartridges were used from Supelco (www.sigmaaldrich.com, Bellefonte, PA, USA), as reference sorbent for enrichment of aflatoxins.

Apparatus and analytical conditions

Transmission electron microscopy (TEM) images of silica materials were obtained using a Jeol (www.jeol.com.jp, Tokyo, Japan) model JEM-1010 microscope operating at 200 kV. A MegaView III camera was used for obtaining images provided with the AnalySIS image data acquisition system.

X-ray diffractograms (XRD) were obtained using Cu K α radiation (Seifert 3000TT θ - θ), and ²⁹Si nuclear magnetic resonance (NMR) spectra were measured with a Varian Unity 300 spectrometer in magic angle spinning (MAS) mode. Porosity data such as pore size and volume, and surface area were determined by using nitrogen adsorption-desorption isotherms, recorded at 77 K with a Micrometrics ASAP2010 automated sorption analyser. The specific surface areas were calculated from the adsorption data in low pressure range using BET model. Fourier transformation infrared (FTIR) spectrometer Bruker Tensor 27 (www.bruker.com, Massachusetts, USA) was also employed. Confocal images were acquired with a Leica TCS-SP2 confocal microscope (www.leica.microsystems.com, Leica Microsystems, Heidelberg GmbH, Germany) at the Central Unit of Research of the University of Valencia equipped with an ultraviolet light

source of continuous wavelength argon excitation laser beam ($\lambda_{\text{exc}} = 365 \text{ nm}$) for aflatoxin ($\lambda_{\text{em}} = 435 \text{ nm}$). The gain was adjusted by using aflatoxin-free UVM-7 pure silica as blank, and later this same gain value was used to record the remaining images. Under these identical conditions, the silica containing aflatoxin leads to emission signal saturation, this indicating in a conclusive way its presence and dispersion.

For SPE procedures, a Vac Elut 20 connected to a vacuum pump system (CKNF model N 0.26.3 AN 18) was employed. Tea samples were previously filtered with polyamide $0.45 \mu\text{m}$ Sartorius Stedim Biotech filters (www.sartorius.com, Goettingen, Germany). Also, a miVac sample concentrator from SP Scientific (www.spscientific.com, Warmister, PA, USA) were used for solvent evaporation.

During optimization stage, aflatoxins were determined using a LC-2000 Plus liquid chromatograph equipped with a FP-2020 Plus Intelligent Fluorescent Detector, a PU-2089 Plus Quaternary Gradient Pump with integrated degasser and an I/FLC/NetII/ADC interface from Jasco (www.jascoinc.com, Madrid, Spain). Injection volume was $20 \mu\text{L}$ and a Kromasil C18 column from Análisis Vínicos (www.analisisvinicos.com, Ciudad Real, Spain) were used for the separation ($150 \times 4.6 \text{ mm}$, $5 \mu\text{m}$ particle size). The separation was carried out at 35°C under an isocratic flow of $1 \text{ mL}\cdot\text{min}^{-1}$, and $\text{H}_2\text{O}:\text{ACN}:\text{MeOH}$ as a mobile phase (65:25:10). The fluorimetric detection was carried out at $\lambda_{\text{ex}} = 365 \text{ nm}$ and $\lambda_{\text{em}} = 435 \text{ nm}$.

On the other hand, for sample analysis an Acquity UPLC Waters liquid chromatograph were used (www.waters.com, Milford, MA, USA), equipped with a triple quadrupole mass spectrometry (MS/MS) detector. In this case a C18 BEH Waters column were used ($2.1 \times 50 \text{ mm}$, $1.7 \mu\text{m}$ particle size), and the separation was carried out at 30°C using a mixture of MeOH and water (containing ammonium formate 0.15 mM and formic acid 0.1%) under gradient conditions (see Table S1). The specific transition ions monitored for each aflatoxin are given in Table S2.

Preparation of mesoporous silica materials

The synthesis of the UVM-7 materials was carried out by a modification of the so-called atrane route [21], through a one-pot surfactant-assisted procedure. This protocol, based on previous works [24, 27], was modified depending on the surfactant extraction procedure. UVM-7 materials were prepared keeping the same molar ratio of the reagents ($2 \text{ Si}:7 \text{ TEA}:0.5 \text{ CTAB}:180 \text{ H}_2\text{O}$) and following similar general procedure. Briefly, 25 mL of TEA were added to 11.2 mL of TEOS and the mixture was warmed at 140°C . Then, the mixture was allowed to cool at room temperature and 4.56 g of CTAB were added to the mixture at 120°C . Subsequently, at 85°C , 180 mL of water were slowly added under

vigorous stirring. This mixture was aged at room temperature overnight and resulting solid was collected by filtration, washed with water and ethanol, and dried with vacuum. Finally, the surfactant was removed by calcination at 550 °C for 5 h, or by acid extraction. In this case, 100 mL of ethanol and 8.5 mL of HCl 37% were added for each gram of the synthesized material, and mixed under vigorous stirring at 60 °C for 12 h.

For the retention study, other three silica materials were prepared for comparative purposes. UVM-11 material was prepared by following procedure described in a previous work of our group [29], and it was carried out in the absence of surfactant. Likewise, Stöber spherical particles, either mesoporous or massive, were synthesized through the well-known Stöber process previously described [30].

Sample pretreatment and SPE protocol

SPE cartridges were prepared by packing the desired amount of UVM-7 material (either calcined or extracted) between two polyethylene frits of 20 µm into 6 mL empty polypropylene cartridges. After optimization, the recommended SPE procedure consisted in using 100 mg of extracted UVM-7 material, which was previously conditioned with 5 mL of MeOH and 5 mL of ultrapure water. Then, spiked ultrapure water containing aflatoxins (AFB₁, AFB₂, AFG₁ and AFG₂) was passed through the cartridge. After that, the cartridge was dried with air for 10 min and analytes were eluted with 3 mL of MeOH. After their concentration by evaporation and reconstitution with 250 µL of MeOH, the extracts were analysed by HPLC-FLD or UHPLC-MS/MS.

The optimization study was carried out based on the recovery obtained from spiked water with AFs, by varying one parameter at time and keeping the other parameters constant. For the application of the method to real samples, green and black tea samples from different origins were purchased, both from commercial bags (containing 1.5 g of tea) and herbalist. In order to find out aflatoxin contamination of the consumable liquid portion of the tea, sample treatment was carried out by immersing tea bags in 250 mL of water at 80 °C during 20 min. In the case of herbalist tea (purchased in bulk), the same amount as in the bags were treated. This liquid portion were homogenised, filtered, and 25 mL of filtrate were analysed according to the protocol described above. Likewise, matrix effect was studied by analysing a triplicate of spiked tea sample (600 ng·L⁻¹ for AFB₁ and AFG₁; 150 ng·L⁻¹ for AFB₂ and AFG₂) following this method.

In addition, results were compared with those obtained with the AOAC reference method [19] using IAC cartridges for aflatoxin extraction.

Results and discussion

Choice of material

As we mentioned above, the ability of silica-based mesoporous material as SPE support for the aflatoxin retention is here investigated. The analytes show molecular diameters comprised between 0.9 and 1.2 nm. The mesoporous materials have pore sizes in the range from 2-3 nm. Hence, they are well suited for analyte retention. Also, the stability of silica materials, their large surface area and their ability to interact through hydrogen bonds give them great utility in this field. Among them, MCM-41 materials are the best-known mesoporous silica materials, even though their large particle size may cause diffusion problems, and reduce the accessible surface area for the analyte retention. On the other hand, in UVM-7 materials (see Fig. 1a), which can be considered as a nanometric version of the MCM-41 silicas, the particle size decreases from the micro to the nano domain. It necessarily implies a similar decrease in the intraparticle mesopore length. This fact combined with the preservation of the mesopore diameter and silica wall thickness, confers to the UVM-7 silicas an improved accessibility for the analytes to the whole internal surface area. Hence, UVM-7 material, was selected as the best option for the retention of aflatoxins.

Regarding to the selectivity of the sorbent, silica materials are able to retain organic analytes with a certain polar character, from an aqueous solution. Therefore, our material is not entirely selective for aflatoxins, this being their main disadvantage in comparison with other sorbents commonly used. However, the UVM-7 morphology, and the possibility of controlling pore sizes through the surfactant-assisted synthesis, gives it a proper structure for the retention of these compounds. Since the retention inside the pore cavities is significantly greater, size-exclusion mechanisms can favour the adsorption of aflatoxin against other compounds. Thus, although some other species may be also retained, the combination of this sorbent with separation techniques such as liquid chromatography with selective detectors (FD or MS/MS) allows the preconcentration and determination of aflatoxins in a complex food matrix.

Optimization of SPE conditions

The comparison between calcined and extracted was firstly carried out, as well as the study of sorbent amount. For this purpose, SPE cartridges were prepared with 25, 50 and 100 mg of each material, and 10 mL of spiked ultrapure water (150 ng L^{-1}) were loaded into each cartridge. After elution with 3 mL of MeOH, the best recoveries were obtained for 100 mg of extracted UVM-7, being these recoveries greater than 90%. Thus,

these cartridges were selected for method development. Additionally, these results are consistent with those obtained in RMN characterization (see below). As it can be seen, extracted UVM-7 shows less Q^4 centres, which means that more silanol groups are available to interact with analytes. It should be noted that the optimization was carried out with AFB₂ and AFG₂, while all analytes were studied for the analytical figures of merit and the analysis of real samples.

Then, loading conditions such as ionic strength and pH were studied. Firstly, the effect of the pH was studied. For this purpose, several spiked solutions were used with different pH values (between 4.5 and 7.09). Results indicated that the analytes recovery were not affected by loading pH, since aflatoxins acid-base activity is expected to be insignificant, and working pH range is not below silica isoelectric point (2-3) [31]. Secondly, the influence of ionic strength was also studied, in order to reduce the solubility of the analytes in the aqueous phase, while increasing their retention into the sorbent. Different contents of NaCl (up to 2 M) in the spiked sample were studied and the results indicated that recoveries were not affected by ionic strength variation, being these observations consistent with other studies using this material as SPE sorbent [27]. Consequently, the use of sodium chloride was discarded.

Even though recoveries obtained with methanol as elution solvent were satisfactory (> 95.9%), other solvents were tested as eluents in order to increase these recoveries. Thus, acetonitrile and ethanol were investigated. For this purpose, 3 mL of each solvent were used as eluent. Low recoveries were obtained with ethanol, whilst using more polar solvents such as MeOH and acetonitrile gave recoveries higher. In fact, best extraction efficiencies were obtained using methanol for both analytes, being also this solvent less toxic than acetonitrile. Subsequently, eluent solvent volume was studied. In this way, volumes from 1 to 4 mL of MeOH were investigated, in order to assure the minimum solvent volume required to elute retained aflatoxins. The greatest extraction efficiencies (103-104%) were obtained when 3 mL of MeOH were used.

In addition, the possibility of increasing the sensitivity of the procedure by evaporating the solvent at 60 °C using rotatory evaporator, were assessed. Then, the residue was redissolved either in 250 µL, 500 µL, 1 mL or 2 mL of MeOH. Results showed that after evaporation of the solvent and its redissolution, no significant differences in recovery were obtained with none of these volumes (variations < 5%), which indicates that it is possible to carry out this preconcentration step.

In order to obtain reliable analytical results and high preconcentration factors, it is relevant to achieve satisfactory recoveries for all analytes in large sample volume. For

the study of the breakthrough volume, volumes between 10 and 100 mL of aflatoxin spiked sample were passed through the SPE cartridge following the procedure described above, and keeping the absolute amount of analytes constant (1.5 ng). Experimental results showed satisfactory recoveries with 50 mL (95.7-98.0%) as well as with 100 mL (97.1-101.2%). For this reason, depending on sample concentration, this material allows to treat up to at least 100 mL of sample in order to achieve the maximum limits established in the legislation.

Finally, the loading capacity of the sorbent was also studied by passing through the cartridge 50 mL of spiked deionized water (concentrations $0.05 \mu\text{g}\cdot\text{L}^{-1}$ to $2 \mu\text{g}\cdot\text{L}^{-1}$). No significant differences were observed in the recoveries of the analytes. This confirms that this material can be used for SPE clean-up within these concentrations.

Analytical figures of merit

The optimized SPE protocol was evaluated in terms of linearity, selectivity and precision. Calibration standards were prepared and injected twice, both in HPLC-FLD and UHPLC-MS equipment in order to determine the sensitivity of the method by using both techniques. The sensibility of the method in terms of limit of detection (LOD) and quantification (LOQ), were calculated with a confidence level of 95%, and following the latest IUPAC recommendations [32]. In both cases, limits were expressed in terms of absolute amount, as well as in tea sample, following the recommended procedure. Linearity range was estimated following these recommendations, considering the LOQ as the lower limit of linearity [32]. As it can be seen in Table 1, LODs were lower than $0.7 \mu\text{g}\cdot\text{kg}^{-1}$ using both systems, and LOQs were below $2.1 \mu\text{g}\cdot\text{kg}^{-1}$. Thus, this method is sufficiently sensitive for quantify the maximum aflatoxin concentration established in the legislation [4, 17].

The precision of the SPE procedure was evaluated by measuring the repeatability (intra- and inter-day) from extractions of spiked water samples, in terms of relative standard deviation (RSD). The intra-day precision was estimated by analysing a triplicate within a day, whereas the inter-day precision was determined with three series of three independent extractions analysed on three different days (Table 2). The SPE procedure showed good precision, with relative standard deviation (RSD) below 5.1%. As it can be seen in Table 2, good extraction efficiencies were achieved (between 96% and 99%) for all studied aflatoxins.

Then, the reusability of extracted UVM-7 cartridges was also evaluated. The results indicate that no significant decrease of the recoveries was observed when the

cartridges were reused until six times. Is important to note that this is a significant advantage over commercial disposable IAC cartridges.

Sample analysis and matrix influence

The recommended SPE procedure using UVM-7 cartridges was applied for the analysis of different tea samples and the evaluation of the matrix effect. As it can be seen in Table 2, extraction efficiencies in tea matrix were slightly affected, being these recoveries above 88% of all aflatoxins. Thus, results indicate that these cartridges can be properly used for the extraction of aflatoxins in tea matrices, although standard addition calibration may be used in order to eliminate the matrix effect. In addition, according to AOAC guidelines [19], recoveries should be over 80% for AFB₁, AFB₂ and AFG₁, and over 60% for AFG₂, in studied concentration range. Thus, the extraction efficiencies are appropriate.

In order to validate the viability of the method using UVM-7 material, SPE protocol were used for the determination of aflatoxins in eight tea samples. These samples were from commercial bags, both green (M1, M2, M3) and black (M4, M5, M6), and also from herbalist (M7 green and M8 black). Standard addition calibration method was applied in order to eliminate the influence of the matrix. As it can be seen in Table 3, no target compounds were founded, except for AFG₂ in samples M1, M4 and M6, being them both green and black tea samples. However, although these values were over the quantification limit, none of them were above the Spanish legislation [17] for total aflatoxin concentration ($10 \mu\text{g}\cdot\text{kg}^{-1}$), except sample M3. Is important to notice that this sample was stored during several years in bad conditions (in contact with atmospheric air and at 25-35 °C) and its consumption is not recommended.

In addition, these positive samples (M1, M4, and M6) were also analysed following the AOAC reference method by using IAC cartridges [19]. As shown in Table 3, it can be concluded that results are comparable among them by using the paired t-test for comparing individual differences [33] for a 95% of confidence level.

Characterization of mesoporous silica-based materials

Firstly, TEM micrographs were obtained for UVM-7 materials, showing the continuous silica network constructed from aggregated small mesoporous nanoparticles (Fig. 2a). Thus, as it can be observed, its framework combines large interparticle non-ordered macropores and intraparticle mesopores showing a hexagonal disordered array (see inset on Fig. 2a). Besides, XRD patterns of both calcined and extracted UVM-7 were taken (Fig. S1), as well as ²⁹Si NMR MAS spectra (Fig. S2). Textural parameters

of the prepared mesoporous materials were also determined from the nitrogen adsorption-desorption isotherms (Fig. S3 and Table S3). In these characterization studies, the UVM-7 morphology was proved. In addition, a greater presence of silanol groups was observed for extracted UVM-7 in comparison with calcined material. The potential of UVM-7 material to retain AFs was also explored. For this purpose, characterization of this silica containing retained aflatoxins was considered. The porosity data shows slightly variations on the textural parameters, thus confirming the ability of the silica to interact with analytes. As shown in Fig. S4, the presence of the aflatoxin in the charged material was also proved by the FTIR spectra as well as by confocal microscopy (Fig. S5). Results confirm not only the retention of the aflatoxins but also its homogeneous distribution along the silica particles at micrometric scale (see Electronic Supporting Material).

Porosity and morphology influence in the aflatoxin retention

Once assumed the beneficial effect of a rich-silanol density at the silica surface, we intend to understand the influence of the porosity on the aflatoxin retention. This aspect is key to interpret the high efficiency as SPE sorbent achieved for the UVM-7 silica with aflatoxins. With this aim, three other silica materials (with different pore systems and morphologies) such as UVM-11 materials [29] and Stöber spherical particles [30] were tested as SPE sorbents. These materials were also characterized by TEM. As it can be seen in Fig. 1 and Fig. 2, UVM-11 material presents a continuous silica network constructed from aggregated small solid nanoparticles. This results in an architecture similar to UVM-7 materials, with the presence of non-ordered large pores and the absence of the mesopores (Fig. 1b), as it can be confirmed in the inset of Fig. 2b. Likewise, spherical structure of Stöber particles were observed, either in the massive silica spheres (Fig. 1d) or with the presence of mesopores (Fig. 1c), depicted in the inset of Fig. 2c.

These materials were also compared in terms of their adsorption-desorption isotherms (Fig. S3) and porosity data. As it can be seen in Table S3, the presence of macropores were confirmed in UVM-11 solid, as well as the availability of mesopores in mesoporous Stöber spheres. It should be noted that the presence of mesopores causes a significant increase in the surface area available for the retention ($19\text{-}976\text{ m}^2\cdot\text{g}^{-1}$). In the same way, the absence of mesopores in UVM-11 material entails a significant surface area decrease in comparison with UVM-7.

Thus, 10 mL of spiked ultrapure water ($150\text{ ng}\cdot\text{L}^{-1}$ of each AF) were loaded into cartridges containing these materials, and analytes were extracted and determined. As

it can be seen in Fig. 3, poor recoveries were obtained using massive Stöber particles due to the absence of any porous cavities, whereas the best retention was achieved with the bimodal porous architecture of UVM-7 material. Regarding to other porous materials, macropores from UVM-11 materials improved the retention of the analytes. However, the presence of mesopores is necessary to overcome the 50% of aflatoxin recovery, being results obtained with mesoporous spherical particles significantly better. In short, the aflatoxin retention increases as follows: massive Stöber < UVM-11 < mesoporous Stöber < UVM-7. Thus, it can be concluded that the presence of intraparticle mesoporous plays a key role in the retention of aflatoxins, due to their appropriate size and shape. The presence of macropores in the UVM-7 give it an optimal architecture for the retention of these compounds. These interparticle textural macropores can play a dual role, as efficient and large routes to favour the diffusion of the analytes and also as an additional pore able to interact with the aflatoxins through hydrogen bonds, thus achieving excellent recoveries.

Taking into account the beneficial role played by the mesopores, it is expected that aflatoxins diffuse relatively well along these pores. In this case, a good size and shape adequacy must be required. The geometry of the aflatoxins was studied (see Electronic Supporting Material) and the approximated molecular size of the different stable conformations was estimated. For this purpose, a conformational search considering a microcanonical ensemble and using stochastic dynamics via Verlet velocity algorithm and Amber force field [34] was performed with the Gabedit program [35]. Then, in order to have an idea of the molecules volume, the longest distance and the perpendicular ones were calculated. As it can be seen in Table S4, molecular diameters in the range 0.9-1.2 nm were obtained for all conformers of these compounds. These observations are consistent with results previously obtained, as this size is appropriate for the molecule adsorption in UVM-7 mesopores with diameters around 2.7 nm (Fig. S6).

Conclusions

In this work UVM-7 mesoporous silica material has been synthesized, and applied to the extraction of aflatoxins in tea samples. After optimization of the extraction protocol, extracted UVM-7 was selected as the best solid phase, and good recoveries were achieved both with spiked water and spiked tea samples. These recoveries are better than those presented by other methods reported in the literature (Table 4), for the extraction of aflatoxins from similar matrices. Likewise, concerning to then LODs, our values were better or comparable than those reported in the scientific literature. These

limits are below those established by Spanish and European legislation [4, 17]. The resulting RSD values also improve the precision of reported methods.

Furthermore, the SPE protocol was applied to the analysis of real tea samples, and the method was validated using IAC cartridges as reference method [19]. Thus, both protocols gave comparable results, being our cartridges (100 mg of sorbent costs less than 1 euro) significantly less expensive than commercial ones (more than 10 euros each cartridge). Also, the reusability of the material was demonstrated up to six times. Hence, the method provided good results by using an inexpensive and reusable material being an alternative to commercial expensive IAC cartridges. In addition, the material permits the multiresidual analysis, thus improving its scope. As a disadvantage, although the purification of the samples is achieved with our material, the selectivity of the extraction can lead to the extraction of other undesired species from the matrix. Nevertheless, the use of selective instrumental technique, such as UHPLC-MS/MS allows to quantify the analytes properly.

Additionally, the retention mechanism of the aflatoxins was studied by testing several silica materials, and the proper structure of the UVM-7 was confirmed, being advisable the presence of both mesopores and macropores for analyte retention.

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Conflict of interest

The authors declare that they have not conflict of interest

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Table 1 Analytical figures of merit of the method combined with HPLC-FLD and UHPLC-MS/MS systems

Analyte	HPLC-FLD					UHPLC-MS/MS				
	Linearity ($\mu\text{g}\cdot\text{L}^{-1}$)	Absolute amount (ng)		Tea samples ($\mu\text{g}\cdot\text{kg}^{-1}$)		Linearity ($\mu\text{g}\cdot\text{L}^{-1}$)	Absolute amount (ng)		Tea samples ($\mu\text{g}\cdot\text{kg}^{-1}$)	
		LOD	LOQ	LOD	LOQ		LOD	LOQ	LOD	LOQ
AFB ₁	-	-	-	-	-	0.8-19.3	0.068	0.207	0.45	1.4
AFB ₂	0.26-4.8	0.021	0.064	0.14	0.43	0.8-4.8	0.067	0.204	0.45	1.4
AFG ₁	-	-	-	-	-	1.0-19.3	0.079	0.240	0.52	1.6
AFG ₂	0.26-4.8	0.021	0.065	0.14	0.43	1.3-4.8	0.105	0.321	0.70	2.1

Table 2 Repeatability, extraction efficiency and matrix influence ($\bar{x} \pm s$) of the SPE protocol

Analyte	Repeatability RSD (%)		Extraction efficiency (%)	
	Intra-day	Inter-day	Water	Tea matrix
AFB ₁	4.5	5.1	96 ± 3	88.3 ± 1.8
AFB ₂	1.7	2.0	98.2 ± 1.2	102 ± 8
AFG ₁	3.8	4.2	96 ± 2	97 ± 3
AFG ₂	1.4	2.3	96.2 ± 1.4	92 ± 9

Table 3 Results for aflatoxin determination in tea samples, and comparison with reference method ($\mu\text{g}\cdot\text{kg}^{-1}$, $\bar{x} \pm s$, n = 3)

Analyte	M1		M2		M3		M4		M5		M6		M7		M8	
	UVM-7	IAC	UVM-7	IAC	UVM-7	IAC	UVM-7	IAC	UVM-7	IAC	UVM-7	IAC	UVM-7	IAC	UVM-7	IAC
AFB ₁	< LOD	< LOD	< LOD	-	< LOD	< LOD	< LOD	-	< LOD	< LOD	< LOD	-	< LOD	< LOD	< LOD	-
AFB ₂	< LOD	< LOD	< LOD	-	< LOD	< LOD	< LOD	-	< LOD	< LOD	< LOD	-	< LOD	< LOD	< LOD	-
AFG ₁	< LOD	< LOD	< LOD	-	(1.0) ^a	< LOD	< LOD	-	(0.6) ^a	< LOD	< LOD	-	(0.5) ^a	< LOD	< LOD	-
AFG ₂	5.0 ± 1.2	4.4 ± 0.4	< LOD	-	13.1 ± 1.7	13.5 ± 1.0	(1.7) ^a	-	5.8 ± 0.7	6.4 ± 0.7	(1.9) ^a	-	7.8 ± 0.4	9.4 ± 0.4	(1.4) ^a	-

^aValues below LOQ

Table 4 Comparison of the method with other reported methods

Sample	Clean-up procedure	Recovery (%)	LOD ($\mu\text{g}\cdot\text{kg}^{-1}$)	LOQ ($\mu\text{g}\cdot\text{kg}^{-1}$)	RSD (%)	Determination technique	Ref.
Tea	UVM-7 SPE	88-102	0.14-0.7	0.43-2.1	1.4-5.1	UHPLC-MS	This work
Black Tea	IAC	75-110	0.2-1	-	-	HPLC-FLD	[10]
Green Tea	QuEChERS-C18	76-115	2-10	5-20	9-25	UHPLC-HRMS	[11]
Medicinal herbs	IAC	64.7-112	0.03	0.1	2.3-8.3	LC-MS/MS	[13]
Medicinal herbs	QuEChERS	68.1-95.6	-	1.2-3.8	0.9-12.7	HPLC-MS/MS	[15]
Medicinal herbs	Oasis HLB	77.6-110	10 ^a	25 ^a	5.0-13.1	LC-MS/MS	[16]

^aThese values are given in terms of total amount of aflatoxin (ng)

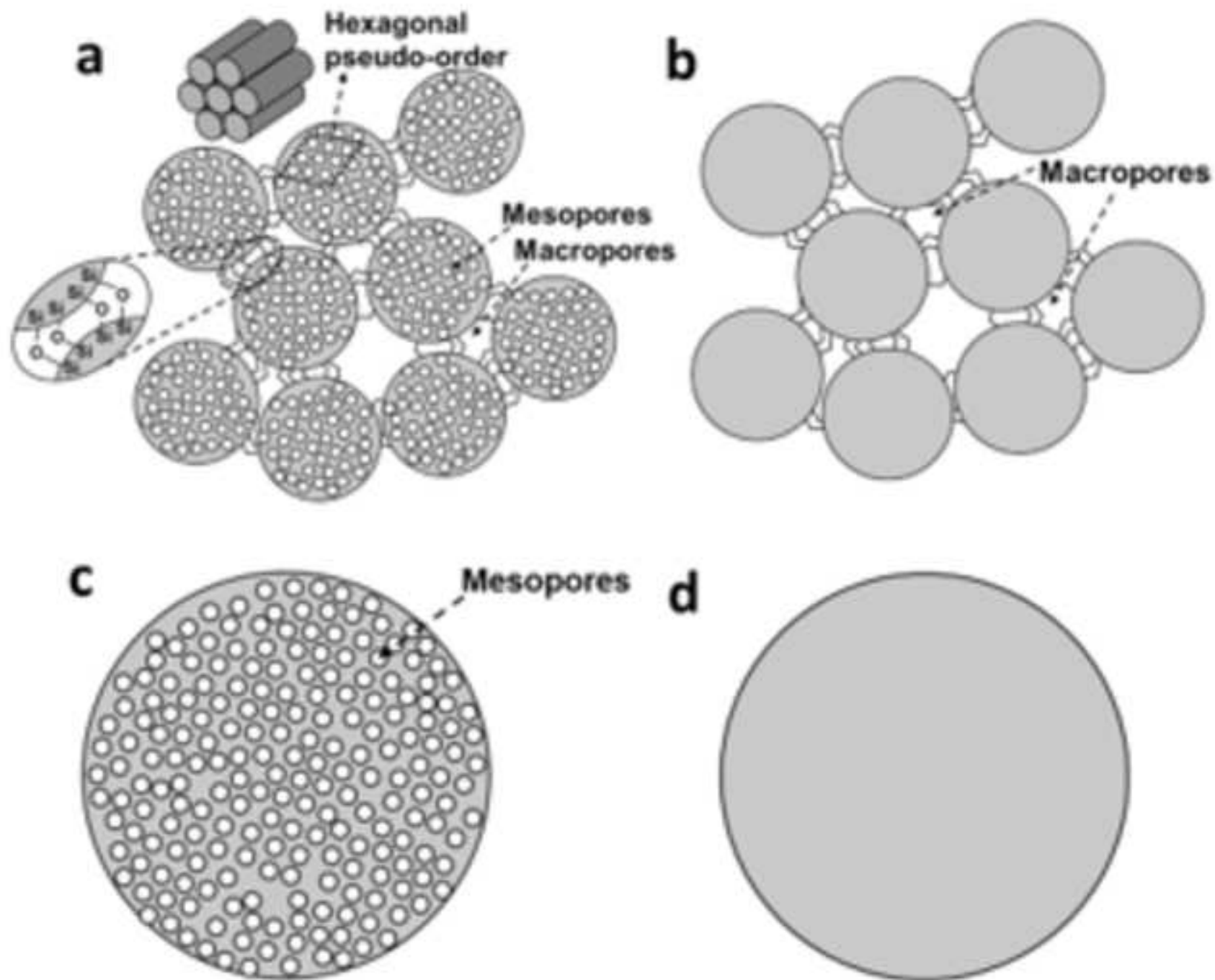
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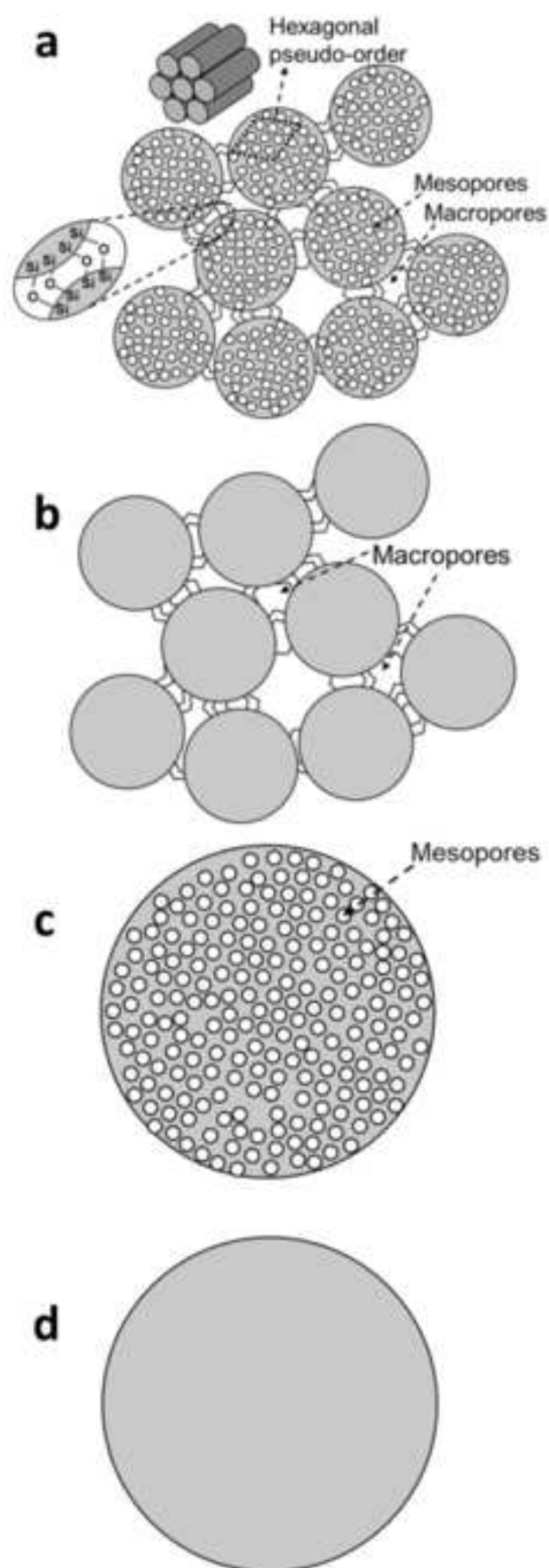
Graphical Abstract Schematic presentation of a mesoporous silica sorbent (type UVM-7) for the extraction of aflatoxins (AF) from tea by solid-phase extraction (SPE), and its determination by liquid chromatography. The morphology of the material allows to retain significantly the analytes

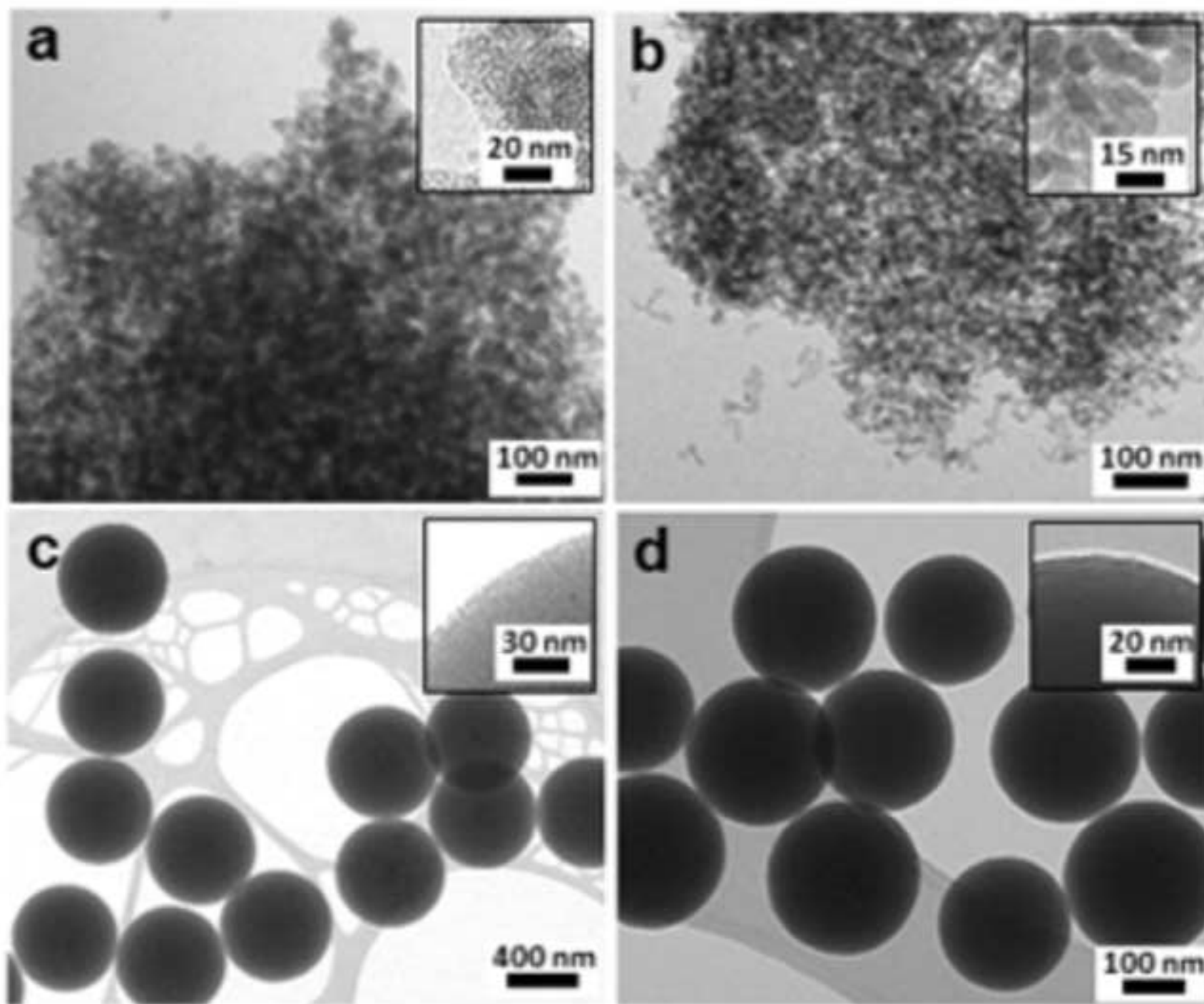
Fig. 1 Schematic representation of the architecture of UVM-7 materials (a) and other silica materials used for comparison study of aflatoxin retention: UVM-11 (b), mesoporous Stöber spherical particles (c) and massive Stöber spherical particles (d)

Fig. 2 Representative TEM images in different zones or regions of UVM-7 materials (a) and other silica materials used for the comparison and study of the aflatoxin retention: UVM-11 (b), mesoporous Stöber spherical particles (c) and massive Stöber spherical particles (d)

Fig. 3 Comparison of the recoveries for aflatoxins B₂ and G₂ by using silica materials with different architectures







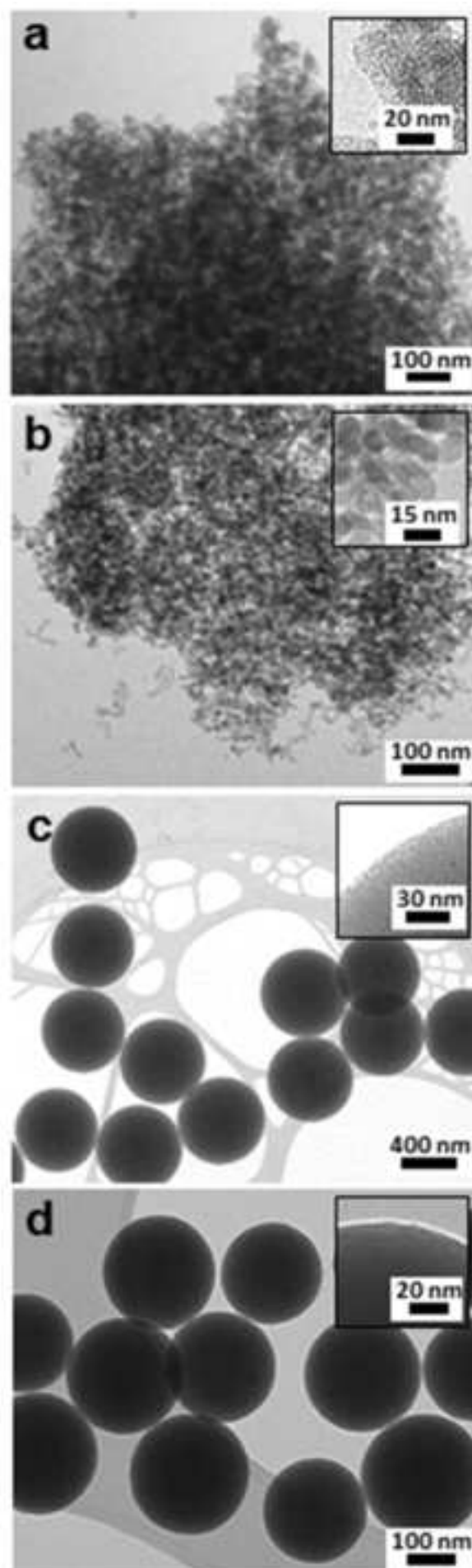


Figure 3

